—GPU-MOC

1

1.1

Moreno Bader Wilson 2024 ParaStell WISTELL-D [1] Liew Ku 1991 CIT [2]

1.2

H. Tsige-Tamirat U. Fischer A. Serikov S. Stickel 2007 McCad ITER 40° 916 CAD [3]

1.3

“ ” —

1.4

Zhou 2025 Kim 2026 online first GPU-MOC [4] [5] MOC 3.1 GPU

2

2.1 RQ1

GPU 1.3

6.1

2.2 RQ2

RQ1

RQ1 6.3 GPU

2.3 RQ3

RQ2 6.3

2.4

1. **GPU-MOC**
2.
3.
4. —

3

3.1

MOC

3.2

Amanatides–Woo 1987 [6] GPU

AMReX [7]

GPU 4

3.3

MOC GPU 9.1

3.4

Gaston 2021 MOC [8]

1.3

5

4 GPU-MOC

4.1

[9]

[10]

4.2

OTF EXP Zhou 2025

EXP OTF 1.5–10

[4]

4.3

Zhou

10% [4 8]

4.4

GPU

[11]

4.5

CPU OpenMOC-OTF C5G7 15 0.9

1640 1 12700KF 1080Ti 2080Ti 3060 4090 32 AMD MI60 6

CPU **300–400** **6** **5.2** **30–100**

[4 1 6 5] EXP/OTF

5

5.1

Baker 2013 GPU-MOC

[12]

5.2

5.3

5.4

AMReX

MOC [7] GPU

5.5

4

5.6

1.3

6 —

6.1

GPU

GPU

1.3

6.2

“ ”

$$|R - R_{ref}| \le \varepsilon_{\text{abs}} + \varepsilon_{rel}|R_{ref}|$$

6.3

6.1

× × ×

Zhou 2025

RQ2 6.1

RQ3

6.5

7.3

6.4

1.3

CUDA

GPU GPU NVIDIA《CUDA C++

Best Practices Guide Release 13.4 9.1 “Timing” [10 §9.1.1] GPU

[10 §9.1.2] CPU GPU

“ — ”

6.5

6.4

6.6

95%

2000

—

1%

—

7.5

6.7

10

GPU

JSON CSV

7

7.1

1.3 $d_{s,j} = T_{,s,j} - T_{,s,j}$

$\Delta T_s = median_j(d_{s,j})$

ΔT_{seq} ΔT

ΔT≥0 6.6 7.3

ΔT=0

7.2

6.3 AMReX

GPU-MOC [7 §5.2]

$|\Delta T_s|$ 5.2

7.4

7.3

6.3 ΔT

7.4

6.6

—

7.4

ΔT=0

7.4

6.1 7.2

6.4 6.5 7.5

6.6

7.5

ΔT<0 6.6

ΔT>0

GPU

GPU

“ ” “

— ”

8

8.1 Zhou 2025

- Zhou 2025 MOC GPU [4]
- GPU EXP OTF Zhou
-

RQ1 RQ2

8.2 Baker 2013

- Baker 2013 [12]

-
- RQ3
  7.5

8.3

–

GPU-MOC

9

9.1

- **MOC** GPU
-
- **GPU**
-
- Python
  C++ Linux Git
- Geant4
  OpenMC Serpent
  GPU-MOC
- **GPU** **MOC** Python C++ GPU
  MOC

9.2

– RQ1—RQ3

-
-
-

9.3

- 7.1
-
- 6.7
-

9.4

RQ1—RQ3 MOC GPU

GPU

5.2 6.6 7.1 RQ3 “

”

10

—— MOC

GPU 9.2

9.4 9.4

—— 6.1

1.3 6.4

—— — 6.3 6.5

6.6

—— 7.3 ΔT 7.4

7.4

——

RQ 9.4 RQ

11 FW-CADIS

11.1

Wagner 2014 [13] FW-CADIS –

11.2

11.3

[1] Moreno C, Bader A, Wilson P. ParaStell: parametric modeling and neutronics support for stellarator fusion power plants[J]. Frontiers in Nuclear Engineering, 2024, 3: 1384788. DOI: 10.3389/fnuen.2024.1384788.

[2] Liew S L, Ku L P. Shielding Design Optimization for the Compact Ignition Tokamak Test Cell Building[J]. Fusion Technology, 1991, 20(2): 164–178. DOI: 10.13182/FST91-A29687.

[3] Tsige-Tamirat H, Fischer U, Serikov A, Stickel S. Automatic generation and validation of an ITER neutronics model from CAD data[J]. Fusion Engineering and Design, 2007, 82(15-24): 1956–1959. DOI: 10.1016/j.fusengdes.2007.06.038.

[4] Zhou F, Li S, Xue R, Bu L, Nie N, Shi P, Wang J, Hu Y, Wang Z, Wang Y, Yang Q, Yu M. Neutron particle transport 3D method of characteristic Multi GPU platform Parallel Computing[EB/OL]. arXiv:2503.17743v1, 2025-03-22 [2026-10-20]. DOI: 10.48550/arXiv.2503.17743.

[5] Kim K M, Im J, Lee H G, Jung Y S. Verification and Performance Assessment of NuDEAL, a GPU-Accelerated Deterministic Transport Framework on Unstructured Meshes[J]. Nuclear Science and Engineering, 2026: 1–25 online first 2026-04-23 . DOI: 10.1080/00295639.2026.2652773.

[6] Amanatides J, Woo A. A Fast Voxel Traversal Algorithm for Ray Tracing[C]. EG 1987—Technical Papers. Eurographics Association, 1987. DOI: 10.2312/egtp.19871000.

[7] Zhang W, Myers A, Gott K, Almgren A, Bell J. AMReX: Block-structured adaptive mesh refinement for multiphysics applications[J]. The International Journal of High Performance Computing Applications, 2021, 35(6): 508–526. DOI: 10.1177/10943420211022811.

[8] Gaston D R, Forget B, Smith K S, Harbour L H, Ridley G K, Giudicelli G G. Method of Characteristics for 3D, Full-Core Neutron Transport on Unstructured Mesh[J]. Nuclear Technology, 2021, 207(7): 931–953. DOI: 10.1080/00295450.2021.1871995.

[9] Boyd W, Shaner S, Li L, Forget B, Smith K. The OpenMOC method of characteristics neutral particle transport code[J]. Annals of Nuclear Energy, 2014, 68: 43–52. DOI: 10.1016/j.anucene.2013.12.012.

[10] NVIDIA Corporation. CUDA C++ Best Practices Guide[EB/OL]. Release 13.4, 2026-10-12[2026-10-20]. DOI

[11] Adinets A. CUDA Pro Tip: Optimized Filtering with Warp-Aggregated Atomics[EB/OL]. NVIDIA Technical Blog, 2014-10-01 2017 11 [2026-10-20]. DOI

[12] Baker C M J, Buchan A G, Pain C C, Tollit B S, Goffin M A, Merton S R, Warner P. Goal based mesh adaptivity for fixed source radiation transport calculations[J]. Annals of Nuclear Energy, 2013, 55: 169–183. DOI: 10.1016/j.anucene.2012.11.029.

[13] Wagner J C, Peplow D E, Mosher S W. FW-CADIS Method for Global and Regional Variance Reduction of Monte Carlo Radiation Transport Calculations[J]. Nuclear Science and Engineering, 2014, 176(1): 37–57. DOI: 10.13182/NSE12-33.